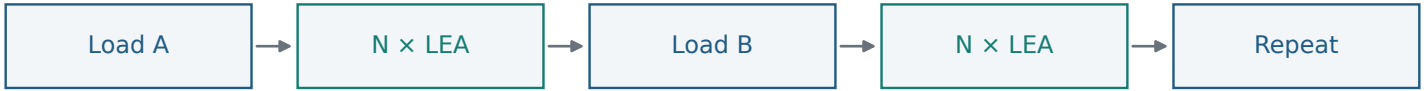
1 Workload and dependencies



LEA (r,r,2) → r computes 3r, updates a 64-bit GPR and does not write FLAGS.

Rotate eight accumulators; compare 4-byte NOPs with equal count/length and serialized loads.

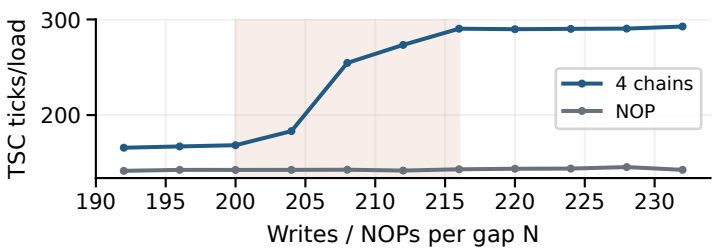
2 Controlled comparison

Scaled LEA, 1/4/8 accumulators; equal-length NOP control; I=10,000; fine step=4.

3 Timed batch and normalization

2 loads and 2N filler writes/loop; ticks/load = elapsed TSC / (2 I).

4 Measured response



5 Inference and control

Four/eight-chain LEA curves rise before the NOP window. This identifies integer-destination pressure under a conditional allocation model. Chain-count and serial-load controls expose scheduler/dependency limits; total physical PRF occupancy is unresolved.